



Name:

Maths Class:

Year 11
Mathematics
Preliminary Course Final Exam
September 2017

Time allowed: 120 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A reference sheet is provided at the rear of this Question Booklet, and may be removed at any time.

Section I Multiple Choice

Questions 1-8

8 Marks

Section II Questions 9-16

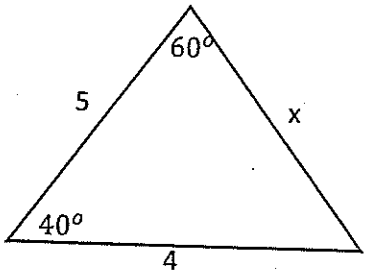
80 Marks

Total = 88 marks

SECTION 1 (10 marks)

Choose the letter corresponding to the correct answer and fill in the Answer sheet provided at the front of your answer booklet.

DO NOT REMOVE THIS SHEET

1	Which of the following is NOT always a true statement? A. The diagonals of a rhombus bisect at right angles B. The opposite angles of a rhombus are equal C. The diagonals of a parallelogram bisect at right angles D. The opposite angles of a parallelogram are equal
2	The quadratic equation $2x^2 - 4x + 5 = 0$ has: A. No real roots B. 1 real root C. 2 equal roots D. 2 distinct Real roots
3	Which statement below is true for the diagram shown?  A. $\cos 60^\circ = \frac{5^2 + 4^2 - x^2}{2 \times 5 \times 4}$ B. $\frac{4}{\sin 60^\circ} = \frac{x}{\sin 100^\circ}$ C. $x^2 = 25 + 16 - 2 \times 5 \times 4 \cos 60^\circ$ D. $\frac{5}{\sin 80^\circ} = \frac{x}{\sin 40^\circ}$
4	Find $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 3}{2x^2 - 5}$ A. $-\frac{3}{5}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. 1

5	If $y = \frac{2\sqrt{3}+3}{\sqrt{3}-2} = x + y\sqrt{3}$, then A. $x = 12$ and $y = 7$ B. $x = -12$ and $y = 7$ C. $x = 12$ and $y = -7$ D. $x = -12$ and $y = -7$
6	If $y = \frac{1}{(5x-1)^2}$ then $\frac{dy}{dx} =$ A. $\frac{-10}{(5x-1)^3}$ B. $\frac{-10}{(5x-1)}$ C. $\frac{-2}{(5x-1)^3}$ D. $\frac{-2}{(5x-1)}$
7	If $\cos \theta = \frac{k}{5}$ for an acute angle θ, then $\tan \theta =$ A. $\frac{\sqrt{25-k^2}}{k}$ B. $\frac{\sqrt{25-k^2}}{5}$ C. $\frac{5}{\sqrt{25-k^2}}$ D. $\frac{k}{\sqrt{25-k^2}}$
8	If $5^{2x-1} = \frac{1}{125}$ then $x =$ A. 13 B. -12 C. -2 D. -1

SECTION 2

Complete all answers in your answer booklet provided

QUESTION 9: (10 Marks)

Marks

- (a) Expand and simplify: $(x + 3)(x^2 - 3x + 9)$ 1
- (b) Solve the equation: $|3x - 4| = 5$ 2
-  (c) What is the size of one of the exterior angles of a regular pentagon? 1
- (d) (i) What are the Domain and Range of the function $f(x) = \sqrt{16 - x^2}$? 2
- (ii) Sketch $y = f(x)$ 2
- (e) Find the equation of the tangent to the curve $y = \frac{1}{4}x^3 - 4$ at the point P (2, -3) 2
- 

QUESTION 10: (10 Marks) Start a new page

Marks

(a) Find the derivatives of:

(i) $y = x^3 + 3x - 1$

1

(ii) $y = (3x - 5)^4$

1

(iii) $y = \frac{2}{x}$

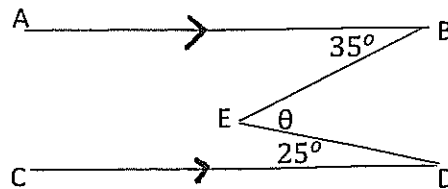
1

(iv) $2\sqrt{x}$

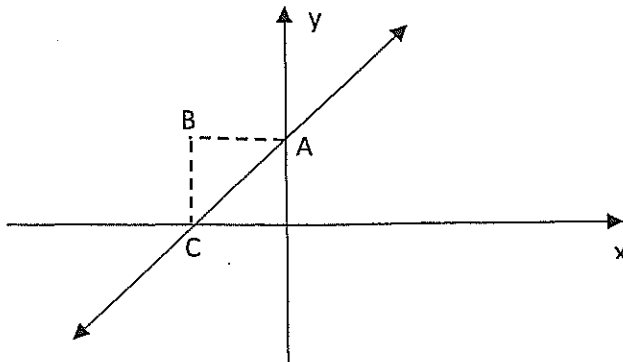
1

(b) Find the size of θ in the following., given $AB \parallel CD$, (no reasons necessary)

1



(c) In the diagram below, the line AC is given as $3x - 2y + 6 = 0$



B has the same x-coordinate as C and the same y-coordinate as A

(i) Find the point B.

2

(ii) Find the equation of the line through B perpendicular to line AC

3

QUESTION 11: (10 marks) *Start a new page*

Marks

- (a) For the function defined by:

$$f(x) = \begin{cases} 2x, & x \geq 1 \\ 2 - 2x, & x < 1 \end{cases}$$

- (i) Sketch $y = f(x)$

3

- (ii) Find the value of $f(-1) + f(1) + f(3)$

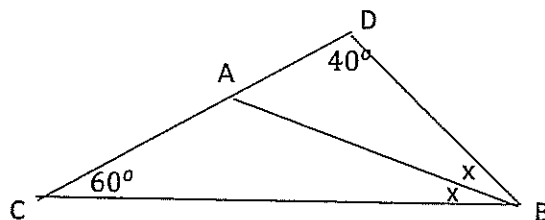
1

- (b) Solve simultaneously $\begin{cases} 4x - y = 19 \\ x + 2y + 2 = 0 \end{cases}$

2

- (c) In the diagram below, AB bisects $\angle DBC$, $\angle ACB = 60^\circ$ and $\angle CDB = 40^\circ$

4



Copy the diagram into your answer booklet
Setting out a formal proof, prove that $\triangle CBA \parallel \triangle CDB$

QUESTION 12: (10 marks) *Start a new page*

Marks

- (a) Find the equation of the normal to the curve $y = 2x^3 - 4x^2$ at the point (1, -2)

3

- (b) α and β are the roots of the quadratic equation $2x^2 - 3x + 5 = 0$
(DO NOT ATTEMPT TO FIND THESE ROOTS)

Find the value of:

- (i) $\alpha + \beta$

1

- (ii) $\alpha\beta$

1

- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

1

- (iv) $\alpha^2 + \beta^2$

2

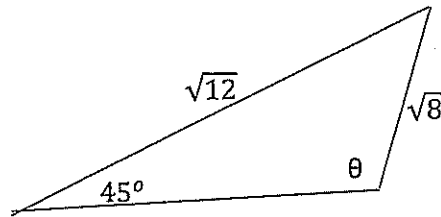
- (v) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

2

QUESTION 13: (10 Marks) Start a new page

Marks
3

(a)



In the diagram above, find the value of θ , if $90^\circ < \theta < 180^\circ$

- (b) (i) On the same diagram shade the region corresponding to the simultaneous solution of:

3

$$(x - 3)^2 + y^2 \leq 4 \quad \text{and} \quad x + y \geq 3$$

- (ii) The point P lies somewhere in the shaded region described in part (i).
At what point in the region above is P furthest from the origin? Give the co-ordinates of this point.

1

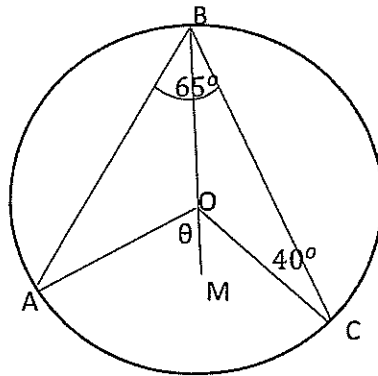
- (c) If the roots of the quadratic equation $kx^2 + (k - 1)x + (2k + 1) = 0$ are such that one root is the reciprocal of the other, find the value of k.

3

QUESTION 14: (10 Marks) Start a new page

Marks

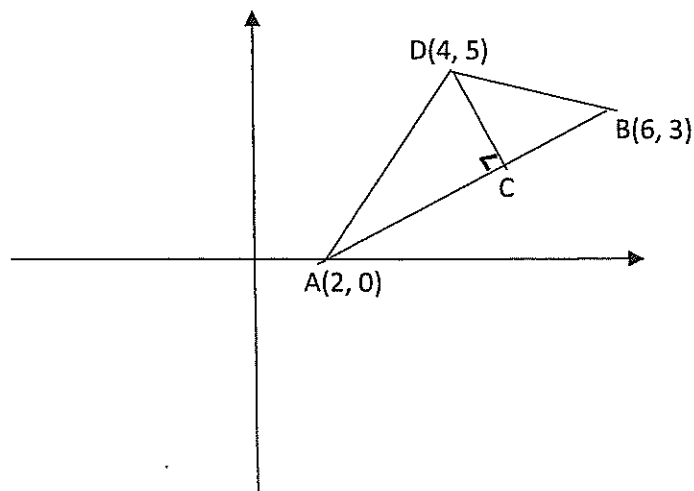
- (a) For the figure below, O is the centre of the circle, $\angle BCO = 40^\circ$
 $\angle ABC = 65^\circ$
 BO is produced to M.



- | | | |
|------|-------------------------------|---|
| (i) | Find the size of $\angle ABM$ | 2 |
| (ii) | Find the size of $\angle AOM$ | 2 |

You must provide reasons for each line of your proofs.

- (b) The point A is (2,0) while B is (6, 3) and D (4, 5) as shown.



- | | | |
|-------|---|---|
| (i) | Find the length of AB | 1 |
| (ii) | Find the equation of the line AB in general form | 2 |
| (iii) | Find the shortest distance of the point D from AB (ie CD) | 2 |
| (iv) | Find the area of $\triangle ABD$ | 1 |

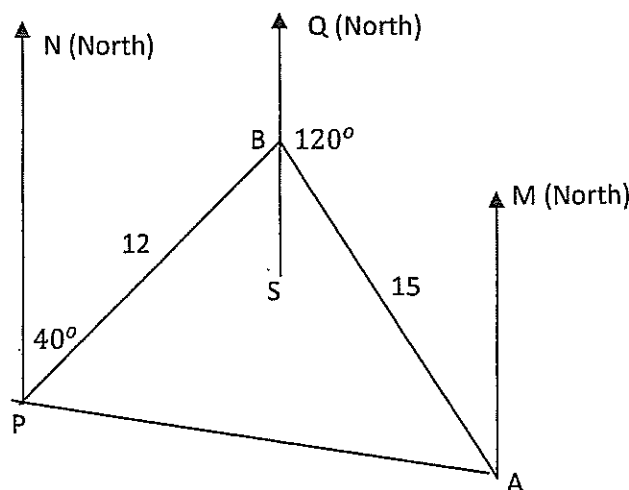
QUESTION 15: (10 Marks) Start a new page

- | | | |
|-----|---|------------|
| (a) | If $f(x) = 3x^2$, find $\frac{f(x+h)-f(x)}{h}$ | Marks
3 |
| (b) | Prove that $\frac{\tan^2 x}{\sec x + 1} = \sec x - 1$ | 2 |
| (c) | Solve $4\sin^2 \theta - 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$ | 3 |
| (d) | If $f(x) = x^{\frac{3}{2}}$ find the value of $f'(4)$ | 2 |

QUESTION 16: (10 Marks) Start a new page

- | | | |
|-----|--------------------------|---|
| (a) | Find $\frac{dy}{dx}$ if: | |
| | (i) $y = \sqrt{x^3 + 3}$ | 2 |
| | (ii) $y = \frac{x}{x+1}$ | 2 |

- (b) The diagram below shows the course of a ship, which sails from a port P on a bearing of 040° for 12 km before changing course to a bearing of 120° and travelling a further 15 km to a destination A.



- | | | |
|-------|---|---|
| (i) | Explain why $\angle PBA = 100^\circ$ | 1 |
| (ii) | Find the distance of A from P to the nearest km. | 2 |
| (iii) | Find the bearing of P from A to the nearest degree. | 3 |

Mathematics Solutions

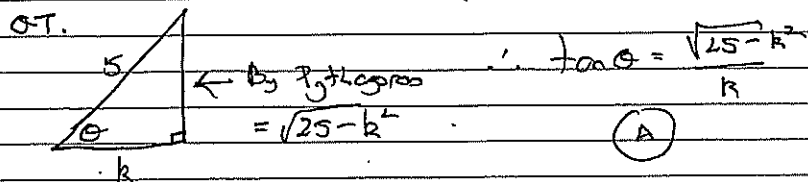
MULTIPLE CHOICE

Q1. (C)

Q2. $\Delta = 16 - 4(2)(5)$
 < 0
 $\therefore$ (A)

Q5. $\frac{(2\sqrt{3}+3)(\sqrt{3}+2)}{3-4} = \frac{4+6+7\sqrt{3}}{-1}$
 $= -12-7\sqrt{3} \therefore$ (D)

Q6. $\frac{d}{dx}(5x-1)^2 = -2(5)(5x-1)^{-3}$
 $= \frac{-10}{(5x-1)^3}$ (A)



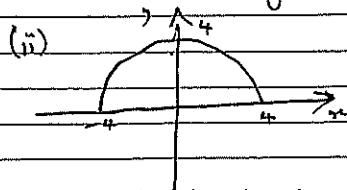
Q8. $5^{2x-1} = 5^{-3}$
 $\therefore 2x-1 = -3$
 $2x = -2$
 $x = -1$ (D)

QUESTION 9:

(a) $x^3 + 27$ (b) $3x = 9$ or $3x = -1$
 $x = 3$ or $x = -\frac{1}{3}$

(c) Sum = 360° $\therefore$ Each angle $\frac{360^\circ}{5} = 72^\circ$

(d) (i) $8 - 4 \leq x \leq 4$
 $0 \leq y \leq 4$



(e) $\frac{dy}{dx} = \frac{3}{4}x^2$
 At $(2, -2)$ $m_T = 3$
 Equation is
 $y+2 = 3(x-2)$
 $y = 3x-8$

Question 10:

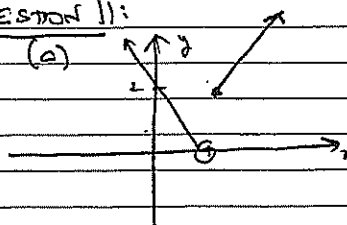
(a) (i) $3x^2+3$ (ii) $12(3x-5)^3$
 (iii) $-\frac{2}{x^2}$ (iv) $x^{-\frac{1}{2}}$ or $\frac{1}{\sqrt{x}}$

(b) $\theta = 60^\circ$

(c) (i) A is $(0, 3)$ X is $(-2, 0) \therefore$ B is $(-2, 3)$

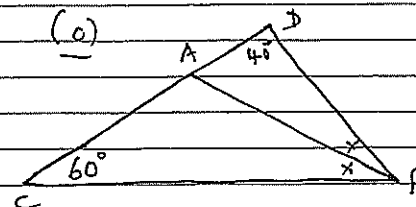
(ii) $m_{AC} = \frac{3}{2}$ m of perp = $-\frac{2}{3}$
 $\therefore$ Equation is
 $y-3 = -\frac{2}{3}(x+2)$
 $3y-9 = -2x-4$
 $2x+3y-5 = 0$

QUESTION 11:



(ii) $f(-1) + f(1) + f(2)$
 $= 4 + 2 + 6$
 $= 12$

(b) $8x-2y = 38$ (1)
 $x+2y+2 = 0$ (2)
 (1)+(2) $9x+2 = 38$
 $\therefore \begin{cases} x = 4 \\ y = -3 \end{cases}$



In $\triangle CDB$, $2x = 80^\circ$ (angle sum)
 $\therefore x = 40^\circ$

In $\triangle CBA$ and $\triangle CDB$
 $\angle ACB = \angle BCD$ (same angle)
 $\angle CBA = \angle CDB = 40^\circ$

$\therefore \triangle CBA \parallel \triangle CDB$ (equiangular)

QUESTION 12:

(a) $\frac{dy}{dx} = 6x - 8$

at (1, -2) $m_T = -2$
 $m_N = \frac{1}{2}$

EQUATION:

$$y + 2 = \frac{1}{2}(x - 1)$$

$$2y + 4 = x - 1$$

$$2y = x - 5$$

(b) (i) $\alpha + \beta = \frac{3}{2}$ (ii) $\alpha\beta = \frac{5}{2}$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ (iv) $\alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \frac{3/2}{5/2} = \frac{3}{5}$$

$$= \frac{9}{4} - 5 = -\frac{11}{4}$$

(v) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$$= \frac{-11/4 + 9/2}{5/2} = -\frac{11}{10}$$

QUESTION 13:

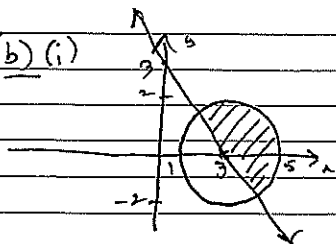
(a) By sine rule, $\frac{\sin \theta}{\sqrt{2}} = \frac{\sin 45^\circ}{\sqrt{3}}$

$$\therefore \sin \theta = \frac{1/\sqrt{2} \cdot \sqrt{2}}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{2}$$

$\therefore \theta = 60^\circ$

(b) (i)



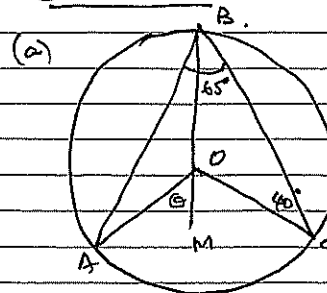
(ii) $P \hat{=} (5, 0)$

(c) Let the roots be α and $1/\alpha$.

$$\text{Product} = 1 = \frac{2k+1}{k}$$

$$\therefore k = -1$$

QUESTION 14:



(i) In $\triangle BOC$, OB and OC are radii.
 $\therefore \angle OBC = 40^\circ$ (isosceles $\triangle BOC$)
 $\therefore \angle BOM = 25^\circ$ (angle sum)

(ii) $\triangle AOB$ is isosceles (equal radii)
 $\therefore \angle BAO = 25^\circ$ (base angles are equal)
 $\therefore \theta = 50^\circ$ (exterior angle of $\triangle ABO$)

(b)

(i) $AB = \sqrt{(6-2)^2 + 3^2} = 5$

(ii) $m_{AB} = \frac{3}{4}$ EQUATION AB: $y = \frac{3}{4}(x-2)$
 $4y = 3x - 6$
 $3x - 4y - 6 = 0$

(iii) $p = \frac{|3(4) - 4(5) - 6|}{5} = \frac{14}{5}$

(iv) Area (ABD) = $\frac{1}{2} (AB)(CD)$
 $= \frac{1}{2} \times \frac{14}{5} \times 5 = 7$ units.

QUESTION 15:

(a) $\frac{f(x+h) - f(x)}{h} = \frac{3(x+h)^2 - 3x^2}{h}$

$$= \frac{6xh + 3h^2}{h} = 6x + 3h$$

(b) $\frac{\tan^2 x (\sec x - 1)}{\sec^2 x - 1} = \frac{\tan^2 x (\sec x - 1)}{\tan^2 x} = \sec x - 1$

(c) $\sin \theta = \pm \frac{\sqrt{3}}{2}$
 $\therefore \theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

(d) $f'(x) = \frac{3}{2}x^{1/2} \Rightarrow f'(4) = \frac{3}{2}(2)$

Question 16:

(a) (i) $\frac{dy}{dx} = \frac{1}{2}(x^3 + 3)^{\frac{1}{2}} \cdot 3x^2$
 $= \frac{3x^2}{2\sqrt{x^3 + 3}}$

(ii) $\frac{dy}{dx} = \frac{(x+1) \cdot 1 - x \cdot 1}{(x+1)^2}$
 $= \frac{1}{(x+1)^2}$

(b) (i) $\angle POS = 40^\circ$ (alternate angles $NP \parallel OS$)
and $\angle SQA = 60^\circ$ (straight QAS)
 $\therefore \angle PBA = 100^\circ$

(ii) In $\triangle PBA$,
 $PA^2 = 12^2 + 15^2 - 2 \times 12 \times 15 \cos 100^\circ$
 $PA \approx 21 \text{ km}$

(iii) $\angle BAm = 60^\circ$ (corresponding angles
 $Q \parallel MA$)
Let $\angle BAP = \theta$
 $\frac{\sin \theta}{12} = \frac{\sin 100^\circ}{21}$

$\therefore \theta = 34.24^\circ$

$\therefore \angle PAm = 94.24^\circ$

$\therefore$ Bearing is $(360 - 94.24)^\circ$
 $= 265.76$
 $= 266^\circ$

EXAMINERS' COMMENTS – YEAR 11 ASSESSMENT 3

MATHEMATICS 2017

QUESTION 9:

d) (i) The Domain was between -4 and 4, not just $x \leq 4$. In most cases, to do the range you really have to know what the graph looks like, and the half-circle gives y as between 0 and 4, not just $y \geq 0$.

QUESTION 10:

- a) Make sure you use the correct notation when differentiating and for each step of working. Students need to be aware of setting out – E.g. $y = x^2 \neq 2x$
For parts (iii) and (iv), those who did not change the original equations to a negative/fractional index first had less success differentiating.
- c) (ii) Some students used the incorrect gradient. Needed to find the gradient perpendicular to line AC. Some found the perpendicular distance and lost 3 marks – Read the question!

QUESTION 11:

- c) If the question asks you to copy the diagram, please do so. Then
 - a. A formal proof requires you to start with “In $\triangle CDB$ and” not just waffle on.
 - b. The only similarity proof involving sides is that two sides are in ratio, about a common angle. SAS is NOT a similarity proof. AA is NOT a reason for similarity. We would accept AAA, but better to say “equiangular”
 - c. For similarity, you only need to prove 2 angles are the same (by angle sum, the third HAS to be the same)
 - d. In this example, x had to be found before progressing.
 - e. Many people used $\triangle CDB$ and $\triangle CAB$ (ok), and then quoted angle ABD which is in neither triangle.

QUESTION 12:

This question was generally well done by all candidates.

QUESTION 13:

- a) You need to mention why $\Theta=60$ is not an answer. The question is worth 3 marks!
- b) ii) Find the point furthest from the origin (0,0) not the centre of the circle.
- c) k is not the root! Since the roots are reciprocal to each other, start with α and $\frac{1}{\alpha}$ then use product of the roots formula.

QUESTION 14:

- a) Students need to give a correct reason for geometry (e.g. the triangle was isosceles as the radii were equal resulting in equal sides)
- b) General form is $ax + by + c = 0$

QUESTION 15:

- a) answer the given question – asked to find an expression not a limit, or a substitution of $x=0$ or $x=h$
- b) convoluted methods used, subject to silly mistakes
- c) forgot the plus/minus sign when solving the equations

QUESTION 16:

- a) (ii) Use the quotient rule when you have a quotient! Do not use the product rule as this is often messy and does not give the simplest form.
- b) (i) “Explain” means “set out with reasons”
(iii) Use $AP = 21$ (from (ii)). You do not need to use more accurate (exact) value for AP since it was found in part (ii)